All About The Back

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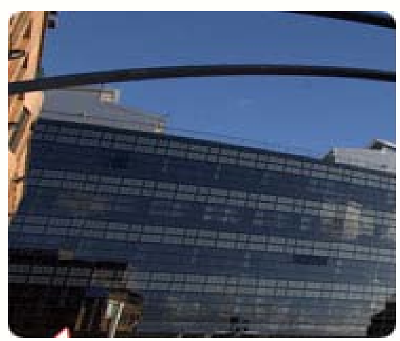
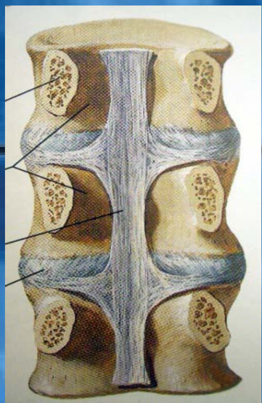
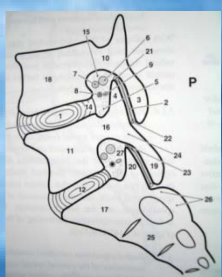
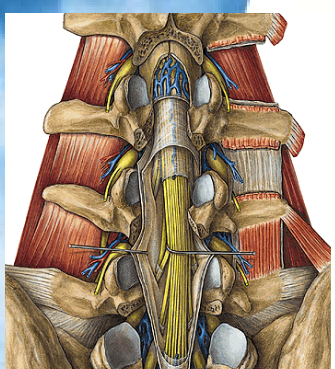


Spine: Biomechanics



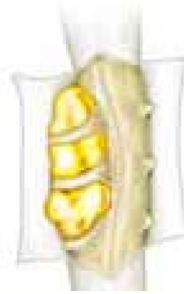


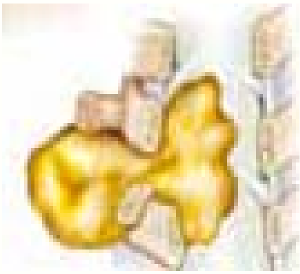


- **Osteo-Liagamentous Complex** (Bones&Co)
  - Crucial for Stabililty
- **FSU (Functional Spinal Unit) 3-Joint Complex**:
  - Discs: Load-bearing (core)+Tensile loads(annulus)
  - 2 Joints: Posterior load path, Compressive Load 10-20%
  - Buckling CSpine 10N / TSpine 20N/ LSpine 88N
- **Muscles:** Superficial / Deep Layer
- **Posture:** CSpine and LSpine Lordosis
  - supine: 200N
  - Standing with 30grd flexion: 1400N

Sources of spinal pain

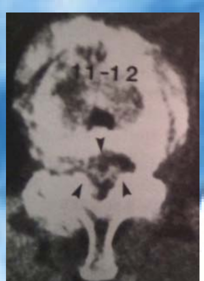
Neoplastic: neuroma, meningioma





Vascular

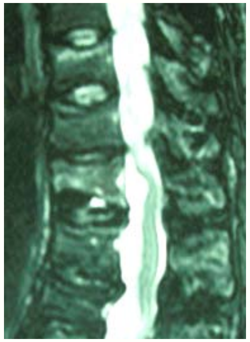
Paraspinal Pathology



Degenerative

Infectious

Osteoporosis



Traumatic

Spine: Sources of Pain in Degenerative Conditions

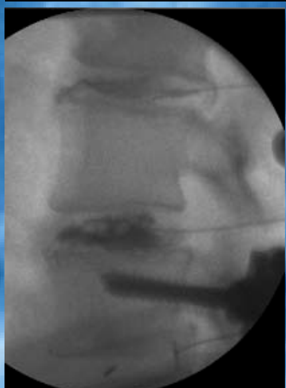
- **Cause of Pain**
  - 19th and mid 20th century: many different aetiologies
  - Since 1950: disc degeneration main driver

- Tendency to prefer **organic reasons**
- Inclination to trust **technical diagnostic** results over clinical judgement

- But: obvious causality and predisposition between **psychiatric disorders** and chronic back pain

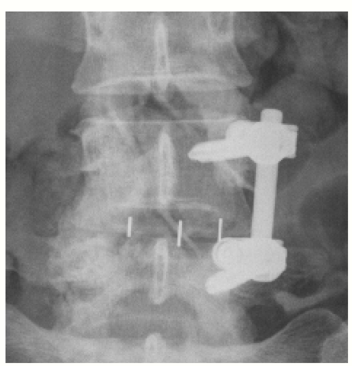
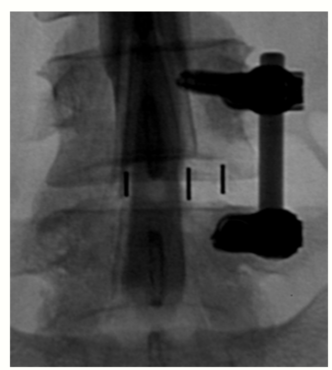
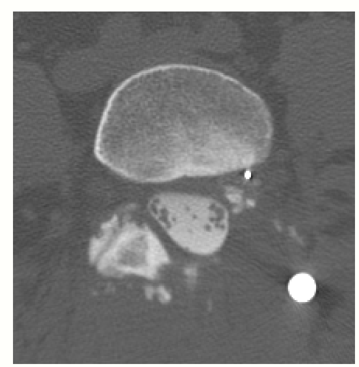
The Mind or The Spine-Which goes first?

- 200 patients
- current and lifetime psychiatric syndromes
  - 77% lifetime criteria
  - 59% current symptoms
    - Major depression: 51% before pain onset
    - Substance abuse: 94%
    - Anxiety disorder: 95%

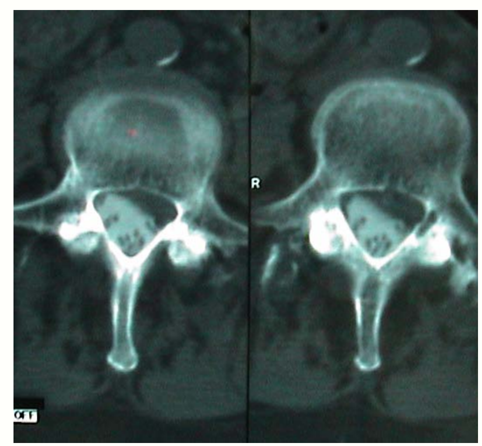




Spine: Establishing the Objective Facts

- XRays + dynamic views (load-bearing!)
- CT -/+Myelography
- MRI -/+Gad +dynamic views
- Provocative Studies
- Diagnostic Blocks
- NCS / EMG

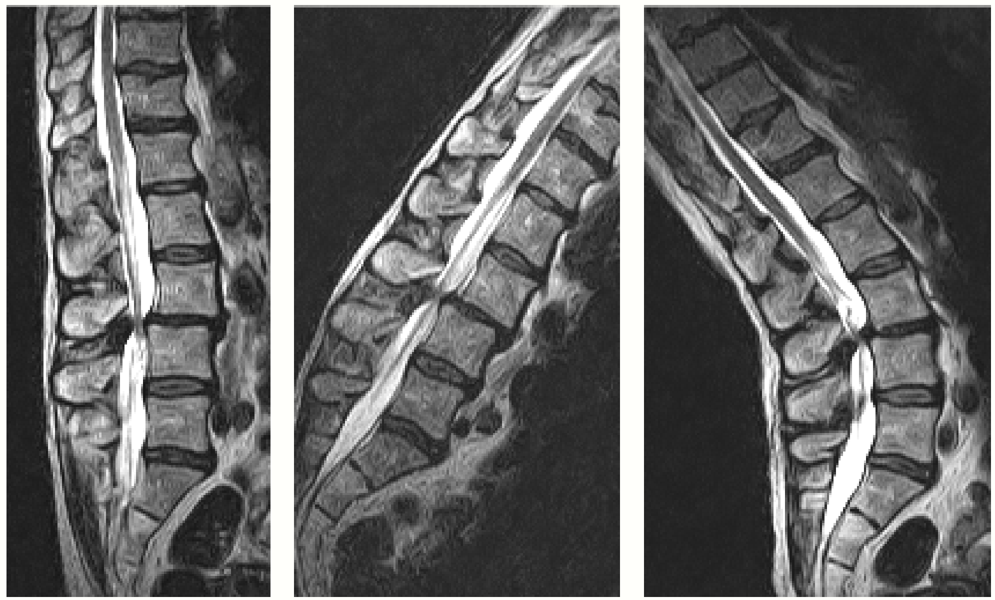





XRay-Myelography and CT-Myelography



Spine: Establishing the Objective Facts
Dynamic MRI

Sit Neutral　Sit Flexion　Sit Extension



BUT:
MRI in ASYMPTOMATIC Subjects

Findings in group of middle-aged people (45 yo average age):

Disc Pathology	**57-64%**
•Disc bulges	39-53%
•Disc protrusion	18-63%
•Disc extrusion	13-18%
•Disc sequestration	0%
•Nerve root compression	4%
DDD (at least one level)	**85%**

Boos et al, Jensen et al, Wood et al

Spine Surgery: Epidemiology & AEtiology

- Lumbar spine interventions /year (US)
  - >250,000 for persistent sciatica
  - >40,000 catheter procedures discogenic pain
  - >300,000 for back pain incldg.
  - >190,000 fusions

- Cost of treatment >170 billions of US$

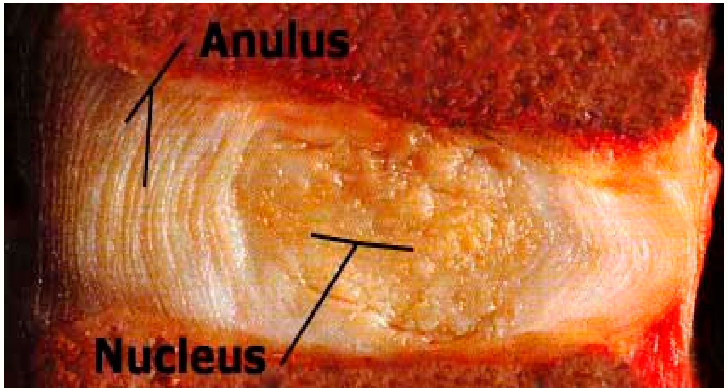
Dewing et al, Spine. 2008;33(1):33-38
Freeman et al., Spine. 2005;30(21):2369-2377
Leigh et al., Arch Intern Med 1997;157-68
National Hospital Survey, 1997. Series 13, No. 144

Degenerative Disorders of the Adult Spine

- **Cervical Degenerative Disorders**
  - Cervical Disc Herniation and Stenosis
  - Cervical Myelopathy
- **Thoracic Degenerative Disorders**
  - Thoracic Disc Herniation and Stenosis
- **Lumbar Degenerative Disorders**
  - Low Back Pain
  - Discogenic Lumbar Pain Syndrome
  - Lumbar Disc Herniation and Stenosis

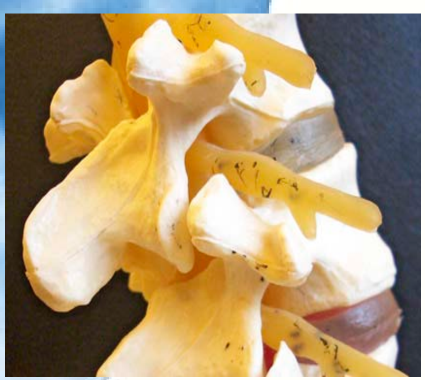
Highest Vulnerability

• Age	40 –49
• Level	L4/5 or L5/S1 (75%)
• Movement pattern	Flexion +Rotation
• Degree of DDD	Stage 2 –Laxity

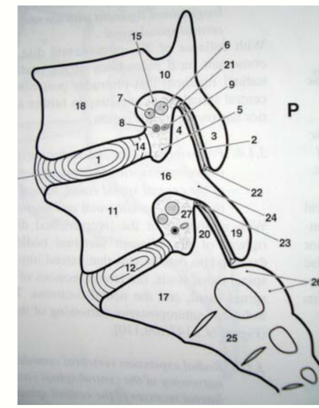


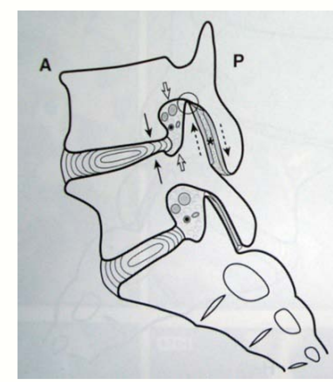
Kirkaldy-Willis (1983) Managing LBP

Common Causes of chronic LBP associated with degenerative disc disease (DDD)



- Intervertebral Disc
- Facet Joints
- Nerve Roots
- Ligaments, Muscles
- SI-Joint





Drawings reproduced from Jinkins et al.

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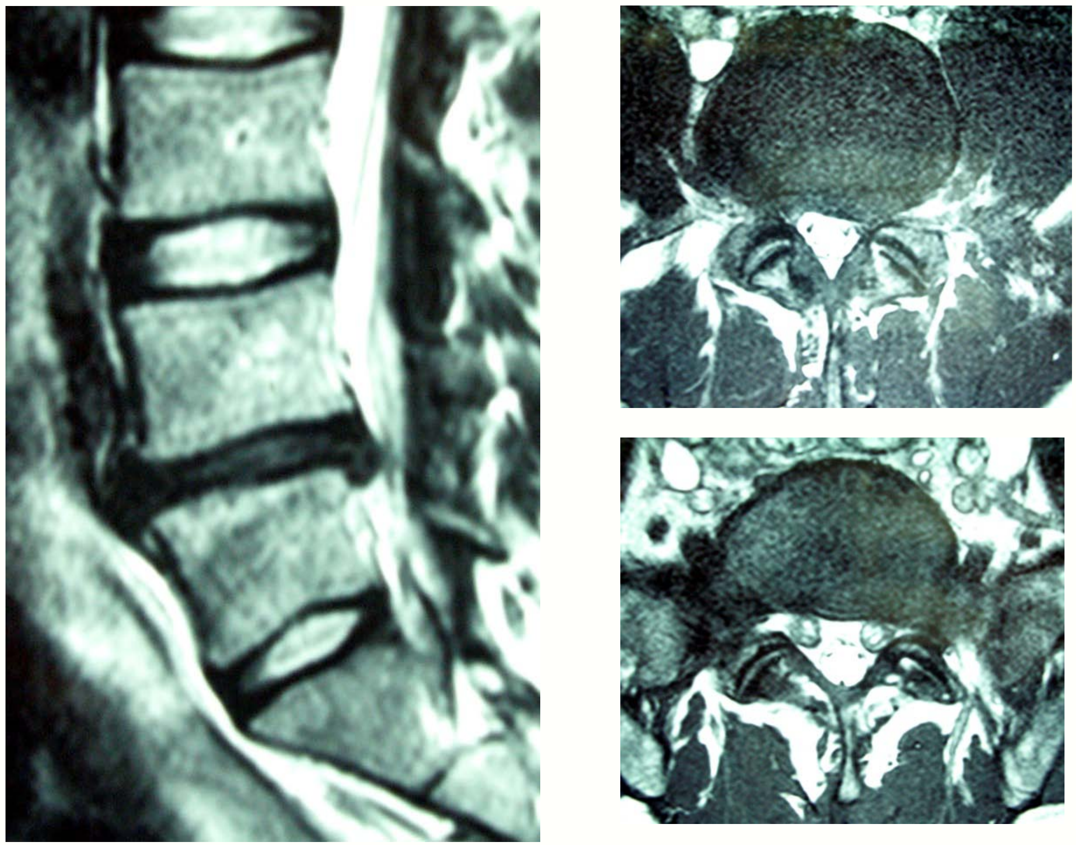
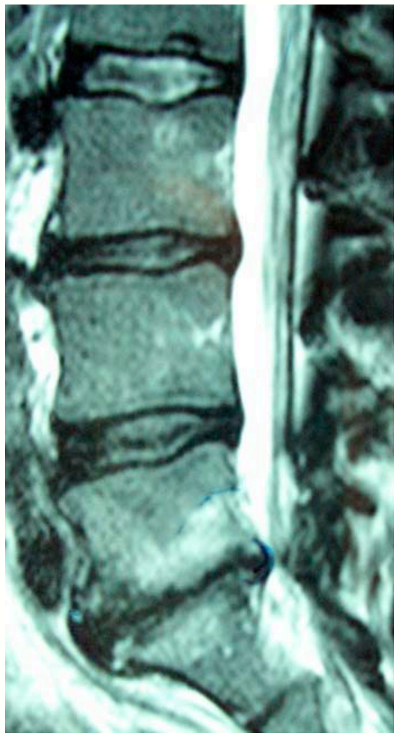
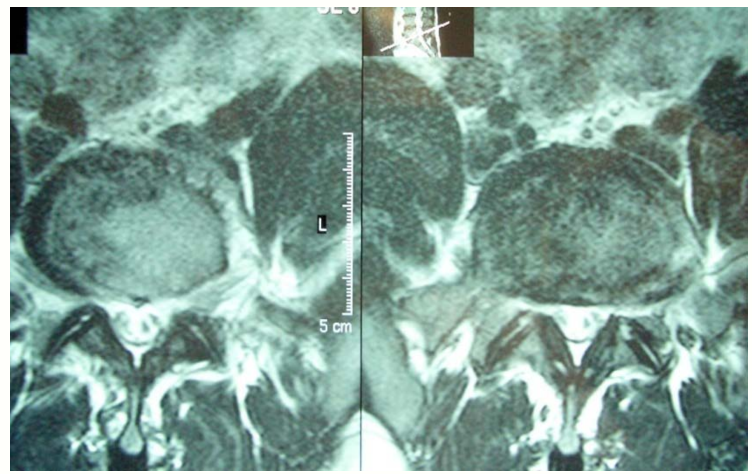
MR Imaging and DDD (moderate)



MR Imaging and DDD (severe)





Clinical Sx of patients with DDD

- Load-dependent LBP
- ± radicular pain
- Pain exacerbated by changing body positions, F/E, rotation, axial load

- Paravertebral spasm most cases
- Anterior tenderness of affected level
- Focal deficits are rare
- ROM is decreased with F/E+lateral bending

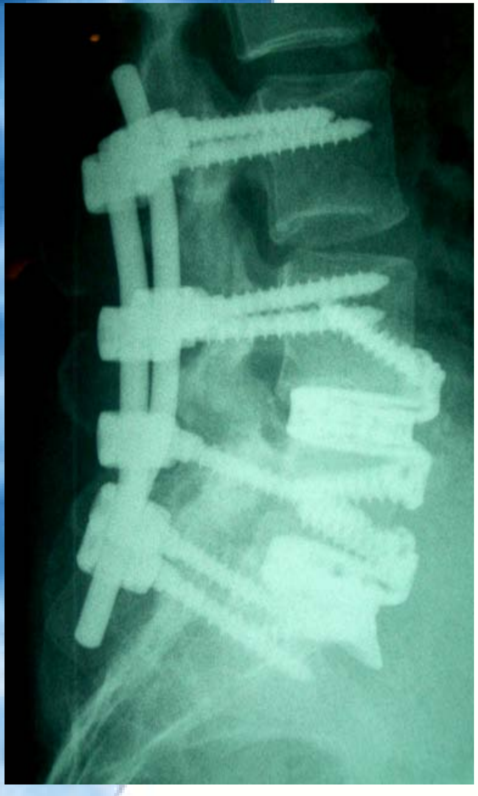
F/E= Flexion / Extension

Differentiating Pain as a Sx

- Radicular Pain (extradural pain)
    - Unilateral
    - Dermatomal in distribution
    - Lancinating, shooting, sharp pain
- Central Pain (intramedullary pain)
    - Deep, poorly defined
    - Intractable
    - No dermatomal relation

Sx Symptom

Elimination of Pain Source can be achieved by:



- Conservative Tx
- Removal of Nociceptor
  - (discectomy, facetectomy, spinal canal decompression)
- Fixation
  - e.g. segmental fusion
- Remodeling
  - Reduction with fusion

Surgical Management of the Spine
Indications for Intervention

- Cauda Equina Syndrome (CES) 1-2% of all pts.
  - retention/incontinence/
  - saddle anaesthesia/erectile d/o
  - outcome: only 50% regain function

- Progressive motor deficit
- Intractable pain
- Instability
- Revision procedure

Annular Tears and Surgical Options

LBP

Discography

Anterior tears only

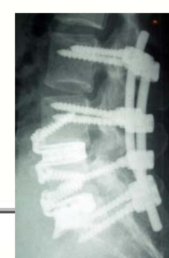
Anterior and Posterior tears

Posterior tears only

TDR

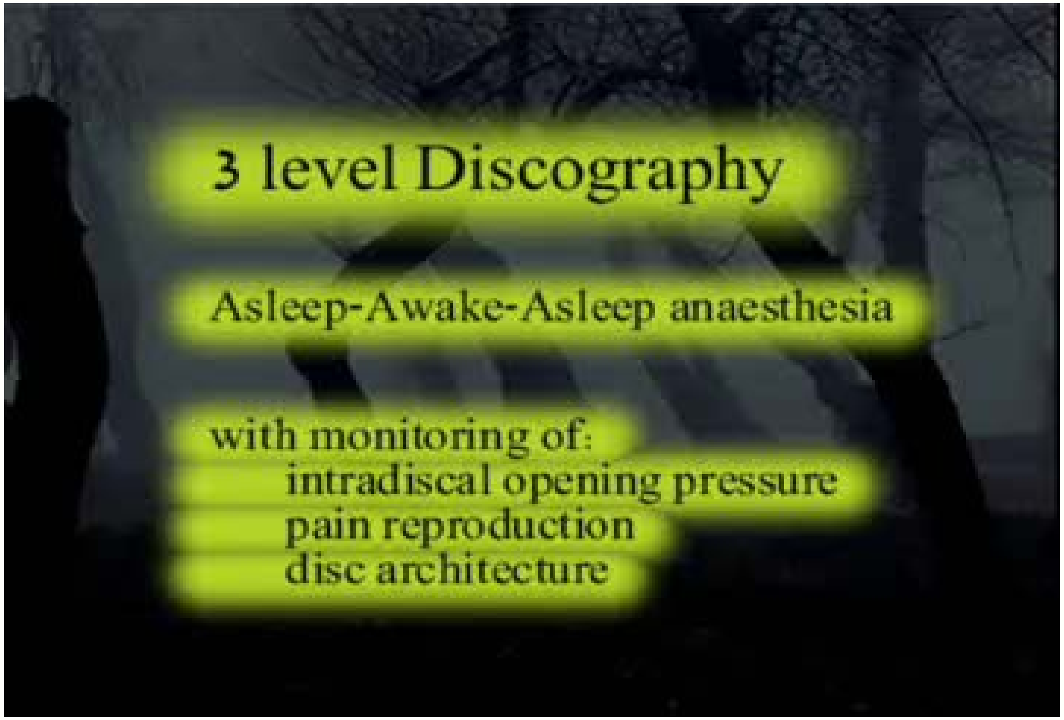
Front-and-Back Fusion / TDR

Dynamic Fusion

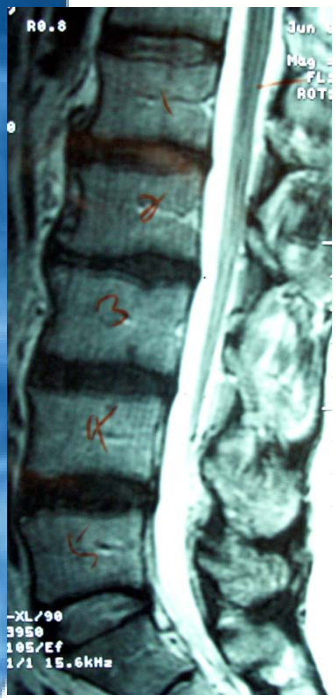


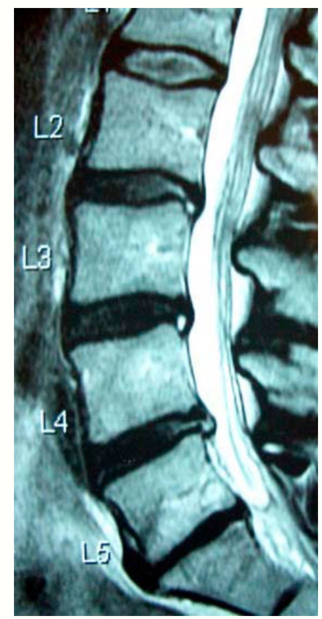
Discography

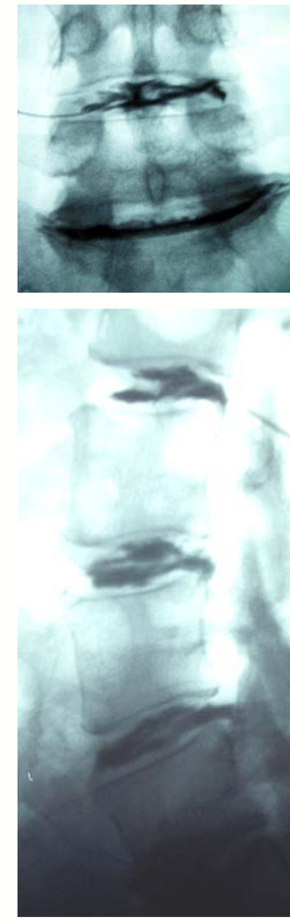
Pressure-monitored Discography
Concordant pain reproduction
Structural changes of the disc



Annular Tears and Imaging







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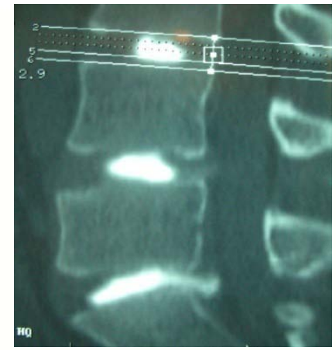
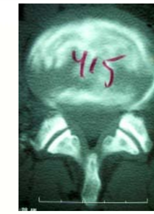
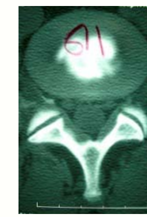
Location of Tears and Therapy

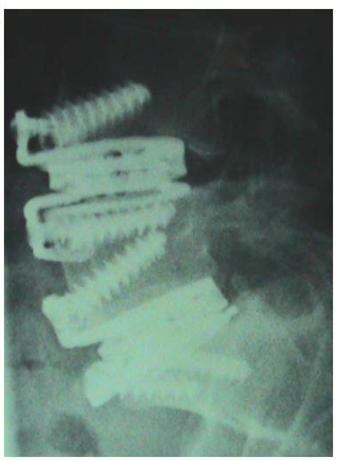
Anterior tear

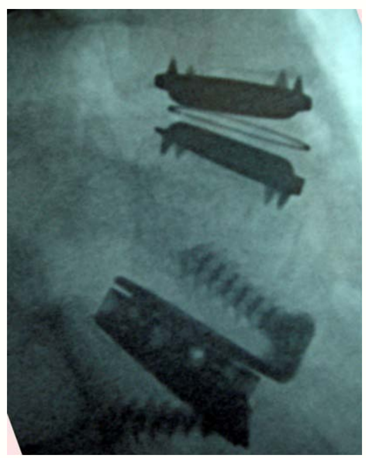




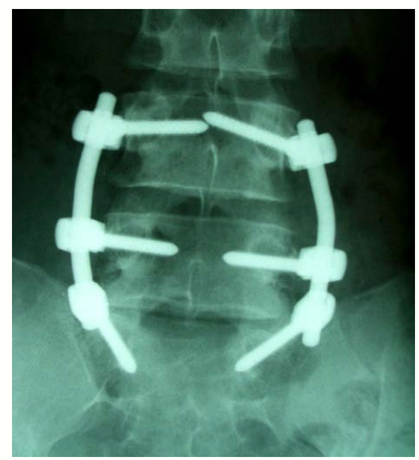


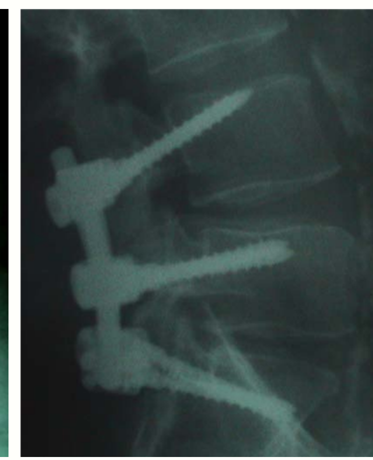
Posterior tear





Topaz/Charite

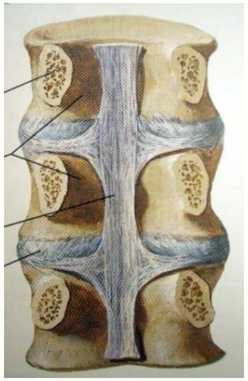




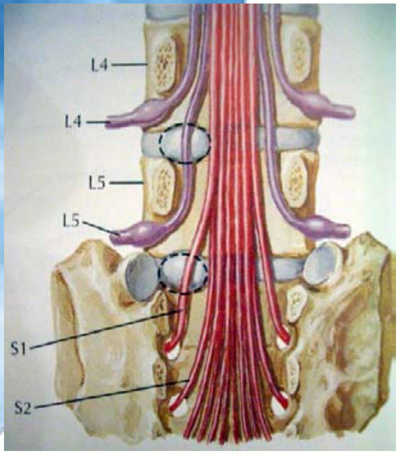
Cosmic

HLD (Herniated Lumbar Disc)

- **Anatomical background**
  - PLL strongest medially > HLD mostly posterolateral



- **Clinical presentation and Findings**
  - LBP +/- radicular pain
  - +/- sphincteric d/o (reduced bladder sensation)
  - (+)ve Valsalva-manoeuvers
  - (claudication) with central HLD
  - Senso-motor deficits >pix
  - SLR (L5-S1), FST (L2-L4), Trendelenburg (L5), FABER (hip)

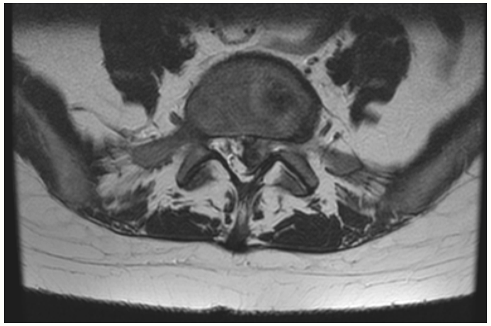


HLD (Herniated Lumbar Disc)
Surgical Treatment

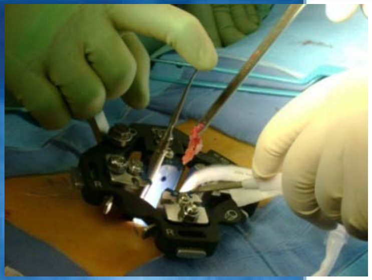
85% of acute HLD improve w/o surgery within 6/52

Surgical Indication:

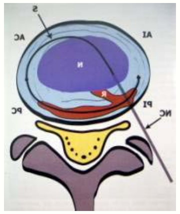
- Cauda Equina Syndrome (CES) 1-2% of all pts.
- retention/incontinence/saddle anaesthesia/erectile d/o
- outcome: only 50% regain function
- progressive motor deficit
- intractable pain

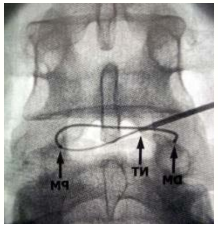


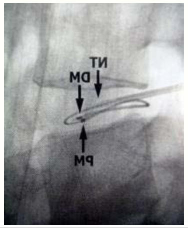
HLD (Herniated Lumbar Disc) Surgical Options



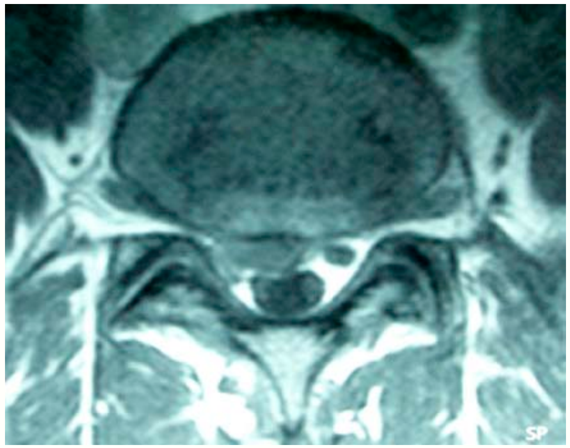
- (Open Discectomy/Laminectomy)
- Microdiscectomy (85% success rate @ 1yr.)
- Endoscopic Procedure
- Intradiscal Procedures
  - Chemonucleolysis (45% @ 1yr.,anaphylaxis)
  - Nucleotomy (35% @ 1yr.)
  - IDET (40% @ 1yr.)

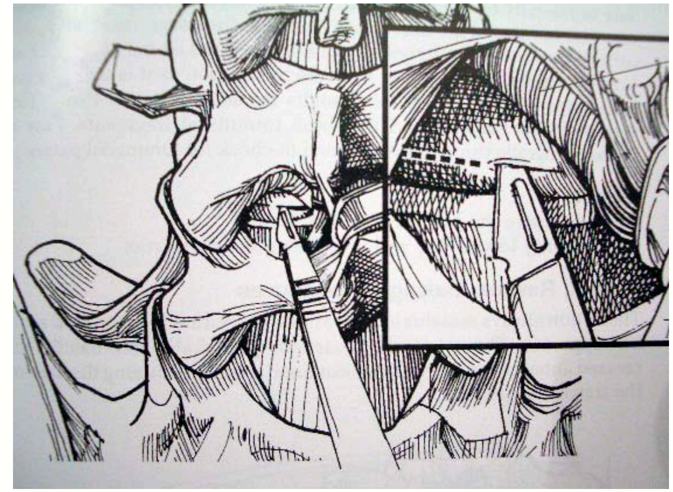


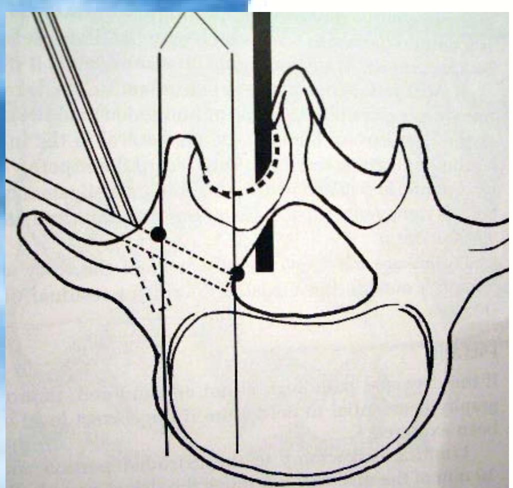


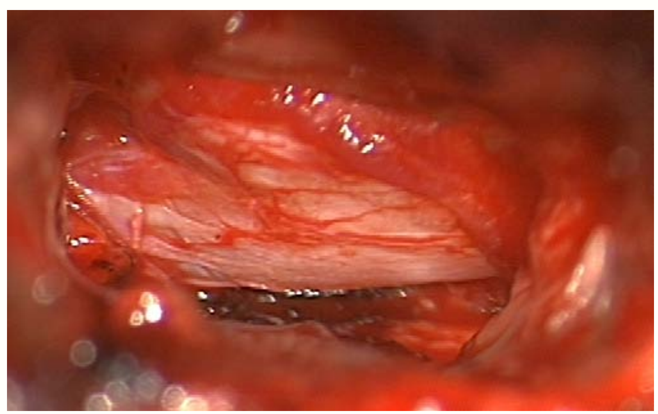


Microdiscectomy









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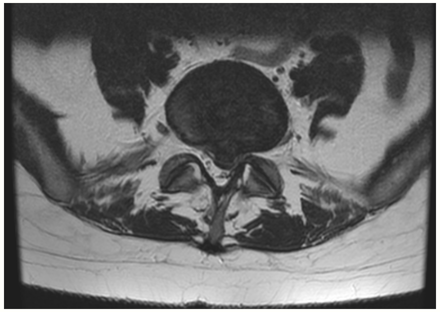
HLD (Herniated Lumbar Disc) Complications

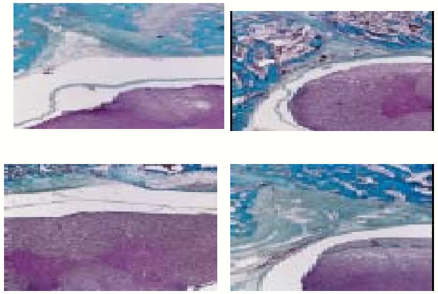


Common

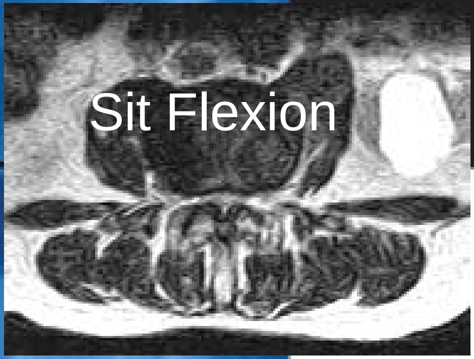
- wound infection 0.5-1%
- motor deficit increased 1-8%
- dural tear CSF leak 0.5-13%
- scar formation (fat graft, barrier gels)

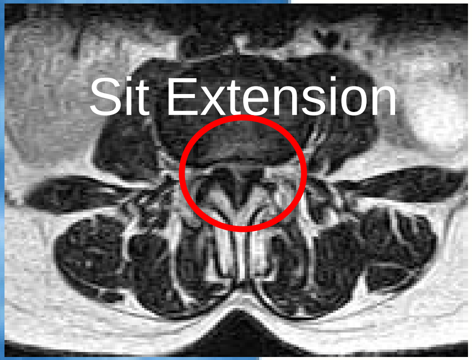
Uncommon

- anterior structure injury (vessels, ureter, bowel, sympathetic..)
- CES 0.2%
- arachnoiditis
- DVT with consecutive PE 0.1%



Spinal Stenosis Treatment





Nonsurgical Management

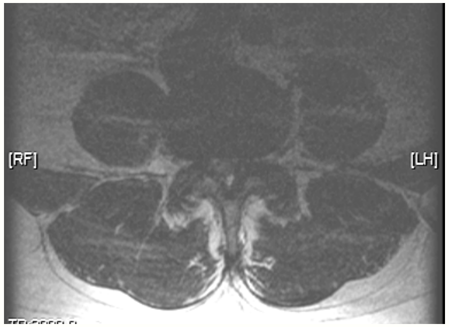
medication

PT +/- traction, general fitness and conditioning

injections (epidurals, DRGB)

TENS

LS-orthosis



Surgical Treatment

is elective (Sx longer >3/12)

is emergent if impeding CES